Aarush Ram Anandh

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12 November 2026

Hiring Team
IBM Research India

Application: Research Intern – AI

Dear Hiring Team,

I am applying for the Research Intern – AI position in IBM Research India's 2027 summer internship program. I am completing a B.Sc. (Hons.) in Computer Science and Mathematics at Ashoka University, expected in 2028. My strongest preparation for this role is hands-on Python/PyTorch research in generative models and reinforcement learning, supported by linear algebra, probability and statistics.

In an ongoing two-person RL coursework project, we co-developed a Gymnasium/poke-env environment with 763 observation features, legality masks over 26 actions, potential-based reward shaping, MaskablePPO and masked DQN agents. We evaluated PPO across two team variants and five seeds each, at 200,000 decision steps per run. Final win rates were 99.7–100% against the fixed heuristic over 1,020 battles per run. Wilson confidence intervals and exact permutation tests found no significant learning-speed difference between variants. This work has given me practical experience with RL training and interpreting controlled experiments beyond a headline score.

At the Constrained Image-Synthesis Lab, my latent-diffusion work achieved $\Delta E_{00} < 1$ and approximately four times better colour accuracy than evaluated open-source baselines, reported in the public preprint *Inference-Time Loss-Guided Colour Preservation in Diffusion Sampling* (arXiv:2601.17259). I am also developing Controlled Generalised Sliced Wasserstein, with a paper submitted to JMLR, C++ computational methods and Lean specifications/proofs. These projects connect mathematical reasoning with research implementation and evaluation.

At Avartan Labs, I doubled rare-class correct detection from 40% to 80% in an 18-class RT-DETR sorting pipeline. At the Koita Centre for Digital Health, I contributed to asthma exacerbation detection achieving approximately 70% sensitivity in study evaluation, and built experimental frozen time-series foundation-model routes that had not yet run in the reported cohort evaluation.

I would welcome the opportunity to bring this implementation and experimental background to IBM's research on language models and agentic systems. I can contribute Python/PyTorch development, RL experimentation and critical analysis of model results while extending my experience to the team's research questions. Thank you for considering my application.

Yours sincerely,

Aarush Ram Anandh